# Mathieu Dagréou

mathieu.dagreou@inria.fr
matdag.github.io
Google Scholar

## Positions

2024-now **Postdoctoral researcher**, *Inria Université Côte d'Azur – Montpellier branch – PreMeDICaL team*, Montpellier, France

2021-2024 **Ph.D. student**, *Inria Saclay - Mind team*, Palaiseau, France

April 2021 - September 2021 **Internship**, *Inria Saclay*, Palaiseau, France
- Advisors: Samuel Vaiter, Thomas Moreau and Pierre Ablin
- Subject: Stochastic bilevel optimization for hyperparameter selection

May 2020 - November 2020 **Internship**, *EDF R&D*, Chatou, France
- Advisors: Alexandre Girard, Yannig Goude, Giorgio Simonini
- Subject: Machine learning for nuclear unit control

## Education & Diplomas

2025 **CNU Qualification for Associate Professor (MCF)**, *Section 26 (Applied Mathematics)*

2021-2024 **Ph.D. in Mathematics & Computer Science**, *Inria Saclay & Université Paris-Saclay*, Palaiseau, France
- Advisors: Samuel Vaiter, Thomas Moreau and Pierre Ablin
- Subject: Contributions to stochastic bilevel optimization

2020-2021 **M.Sc. Mathematics, Vision, Learning**, *École Normale Supérieure Paris-Saclay*, Gif-Sur-Yvette, France

2017-2020 **Engineering degree**, *École Centrale de Nantes*, Nantes, France

2014-2017 **Classes préparatoires**, *Lycée Michel Montaigne*, Bordeaux, France

## Publications

### International Conferences

1. L. Bleistein*, **M. Dagréou***, F. Andrade*, T. Boudou, and A. Bellet. *Optimal Transport under Group Fairness Constraints*. In *International Conference on Machine Learning (ICML)*, **Spotlight (Top 2.2%)**, 2026.
2. **M. Dagréou**, T. Moreau, S. Vaiter, P. Ablin. A Lower Bound and a Near-Optimal Algorithm for Bilevel Empirical Risk Minimization. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
3. **M. Dagréou**, P. Ablin, S. Vaiter, T. Moreau. A framework for bilevel optimization that enables stochastic and global variance reduction algorithms. In *Advances in Neural Information Processing Systems (NeurIPS)*, **Oral equivalent paper (Top 2%)**, 2022.
4. T. Moreau, M. Massias, A. Gramfort, P. Ablin, P.-A. Bannier, B. Charlier, **M. Dagréou**, T. Dupre la Tour, G. Durif, C. F Dantas, Q. Klopfenstein, J. Larsson, E. Lai, T. Lefort, B. Malézieux, B. Moufad, B. T Nguyen, A. Rakotomamonjy, Z. Ramzi, J. Salmon, S. Vaiter. Benchopt: Reproducible, efficient and collaborative optimization benchmarks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

### Preprint

1. **M. Dagréou**, V. Dorseuil, M. Yaghini, N. Papernot and A. Bellet. *Detectability without Interference: Efficient Canary Design for Privacy Auditing in One Run*. arXiv preprint arXiv:2605.27292, 2026.

### National Conferences

1. **M. Dagréou**, T. Moreau, S. Vaiter, P. Ablin. Borne inférieure de complexité et algorithme quasi-optimal pour la minimisation de risque empirique bi-niveaux. In *XXIXème Colloque Francophone de Traitement du Signal et des Images GRETSI*, 2023.
2. **M. Dagréou**, P. Ablin, S. Vaiter, T. Moreau. Algorithmes stochastiques et réduction de variance grâce à un nouveau cadre pour l'optimisation bi-niveaux. In *XXVIIIème Colloque Francophone de Traitement du Signal et des Images GRETSI*, 2022.

### Workshop paper

1. L. Bleistein, **M. Dagréou**, F. Andrade, T. Boudou, and A. Bellet. *Optimal Transport under Group Fairness Constraints*. EurIPS workshop *Unifying Perspectives on Learning Biases*. 2025, `https://openreview.net/forum?id=FAi7sNLS9w`

### Miscellaneous

1. **M. Dagréou**, P. Ablin, S. Vaiter, and T. Moreau. How to compute Hessian-vector products?, In *ICLR blogpost track*, **Highlight (top 10%)**, 2024, `https://iclr-blogposts.github.io/2024/blog/bench-hvp/`

## Awards

2026 **Gold Reviewer**, *ICML 2026*, (Top 25%), `https://icml.cc/Conferences/2026/ProgramCommittee`

2023 **Top Reviewer**, *NeurIPS 2023*, (Top 10%), `https://neurips.cc/Conferences/2023/ProgramCommittee`

2023 **TICS Doctoral School of Paris-Saclay prize**, *https://www.universite-paris-saclay.fr/ecoles-doctorales/sciences-et-technologies-de-linformation-et-de-la-communication-stic*

## Teaching

2023 **Optimization**, *CentraleSupelec*, Teaching assistant

- 2nd year of engineering program at CentraleSupelec, Master's level (M1)
- Supervision of tutorial sessions and practical labs on Matlab, grading of assignments

## Academic service

**Organization** **Workflow Chair**, *AISTATS 2027*, `https://virtual.aistats.org/virtual/2027/organizers`

- Management of the review process on OpenReview for ∼4,700 submissions, ∼3,300 reviewers, 275 Area Chairs and 48 Senior Area Chairs
- Maintenance of the conference website
- Shared role with Jean-Baptiste Fermanian, in close collaboration with the Program Chairs, Quentin Berthet and Aymeric Dieuleveut

**Peer review** **Journals**

- Transactions on Machine Learning Research (2025)
- Journal of Machine Learning Research (2023, 2024, 2025)
- IEEE Signal Processing Magazine (2023)
- Machine Learning (2022)

**International Conferences**

- International Conference on Learning Representations (2026, 2027)
- Neural Information Processing Systems (2023, 2024, 2025, 2026)
- International Conference on Machine Learning (2023, 2024, 2025, 2026)
- Conference on Artificial Intelligence and Statistics (2024, 2026)

**Workshops and national conferences**

- Privacy-Preserving Machine Learning Workshop at EurIPS 2025 (2025)
- Conférence sur l'Apprentissage automatique (2026)

## Communication

2026-07 Talk at Conférence sur l'Apprentissage automatique (Montpellier): *Optimal Transport under Group Fairness Constraints*

2026-03 Talk at Institut de Mathématiques de Toulouse (Toulouse): *Bilevel optimization for machine learning*

2026-01 Talk at Institut de Mathématiques de Bordeaux (Bordeaux): *Bilevel optimization for machine learning*

2026-01 Talk at MAGNET team seminar (Lille): *Bilevel optimization for machine learning*

2025-02 Talk at Machine Learning in Montpellier, Theory & Practice (Montpellier): *Bilevel optimization for machine learning*

2024-06 Talk at STIC doctoral day (Gif-sur-Yvette): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*

2024-05 Poster Session at AISTATS (Valencia): *A lower bound and a near-optimal algorithm for bilevel empirical risk minimization*

2023-09 Poster Session at GRETSI (Grenoble): *A lower bound and a near-optimal algorithm for bilevel empirical risk minimization*

2023-06 Poster Session at the workshop "Optimization and machine learning" (Toulouse): *A lower bound and a near-optimal algorithm for bilevel empirical risk minimization*

2023-02 Talk at Center of Data Science (ENS): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*

2022-12 Poster Session at NeurIPS (New Orleans): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*

2022-11 Poster Session at NeurIPS@Paris (Paris): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*

2022-10 Poster Session at GDR MOA (Nice): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*

2022-09 Poster Session at GRETSI (Nancy): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*

2022-06 Poster Session at Curves and Surfaces (Arcachon): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*

2022-04 Talk at the Parietal Meeting: *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*

2022-03 Talk at Proba-Stat seminar (LJAD Nice): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*

2022-03 Talk at the Miles team seminar (LAMSADE): *A framework for bilevel optimization that enables stochastic and global variance reduction algorithms*

## Computer skills

Language Python, Matlab, R

Tools Unix, Git, LaTeX, SLURM, OAR

## Languages

French Native speaker

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|---|---|
| English | Fluent |
| Spanish | Fluent |